Layer-averaged expectations with a linear-in-height mean profile: Gaussian subgrid variability, quadrature design, and a pill marginal

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Abstract

We study the numerical computation of

$$\overline{G} = \frac{1}{H}\int_{-H/2}^{H/2} \mathbb{E}\big[g(\mathbf{X}(z))\big]\,dz, \qquad \mathbf{X}(z) \sim \mathcal{N}(\boldsymbol{\mu} + z\mathbf{d}, \Sigma),$$

where $\mathbf{X}(z) \in \mathbb{R}^2$ is a subgrid state with **linear** layer-mean profile in a coordinate z (layer depth H), homogeneous Gaussian fluctuations Σ, and a possibly nonlinear, possibly non-smooth response g. The note is **model-agnostic**: the same statistics appear whenever a coarse column carries a linear trend plus multivariate Gaussian noise and one seeks the layer average of $\mathbb{E}[g(\mathbf{X}) \mid z]$. We explain *which* Gaussian quadrature rule matches *which* measure (Gauss–Legendre for uniform z; Gauss–Hermite for Gaussian marginals; inverse-CDF Gauss–Legendre for non-Gaussian laws such as a Gaussian–uniform convolution), derive the tensor-product structure of the natural $\mathcal{O}(N^3)$ reference discretization (*column-tensor*), clarify why "Legendre in z only" cannot replace the inner expectations, and give a Rosenblatt-conditioned $\mathcal{O}(N^2)$ scheme (*profile–Rosenblatt*) whose inner axis is the *pill* distribution. We justify scalar root-finding for quantiles as a **numerical design choice** (not part of the stochastic model), record Brent as a reference inverter, and summarize Chebyshev/rational surrogates with an operation-count comparison. A moist example—(T, q) with saturation excess—is used only for illustration; an appendix records one fixed q_{sat} closure for reproducibility.

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1 Introduction

Coarse grid columns in geophysical and engineering models often carry **subgrid variability**: the true fields fluctuate inside a control volume while only cell-mean (or few-moment) information is evolved. If a closure specifies that fluctuations are approximately **Gaussian** with covariance Σ and that the **conditional mean** of the state varies **linearly** across a layer $z \in [-H/2, H/2]$, then at each height the state is $\mathbf{X}(z) \sim \mathcal{N}(\boldsymbol{\mu} + z\mathbf{d}, \Sigma)$. Many outputs are **nonlinear** functionals $g(\mathbf{X})$—saturation indicators, reaction rates, thresholds—for which $\mathbb{E}[g(\mathbf{X})] \neq g(\mathbb{E}[\mathbf{X}])$ (failure of naive moment closures is often framed via Jensen's inequality when g is convex or concave on relevant domains).

The quantity of interest. The layer-mean statistic is the **double** integral over z and over subgrid fluctuations,

$$\overline{G} = \frac{1}{H} \int_{-H/2}^{H/2} \left(\int_{\mathbb{R}^2} g(\mathbf{x})\, \phi_z(\mathbf{x})\, d\mathbf{x} \right) dz, \qquad \phi_z = \mathcal{N}(\boldsymbol{\mu} + z\mathbf{d}, \Sigma). \tag{1}$$

Equation (1) is the single object that the four schemes below (*center-only*, *moment-matched*, *column-tensor*, *profile–Rosenblatt*) approximate. It is *not* the expectation under a single bivariate Gaussian unless one collapses the z structure (*center-only*, *moment-matched*) or reparameterizes the measure (*profile–Rosenblatt*).

Why quadrature (and which kind) is subtle. The outer average is with respect to **Lebesgue measure** on a **compact** interval; the inner integral is with respect to a **Gaussian** on $\mathbb{R}^2$ at each z. Gauss–Legendre (GL) and Gauss–Hermite (GH) rules are not interchangeable names for "high-order integration": each is Gaussian quadrature for a **specific weight** on a specific domain, hence exact for a **specific** class of polynomials in the correct variables [2, 3]. Using GH nodes as if the marginal along an axis were standard normal when it is actually a **pill** (Gaussian convolved with a uniform) misplaces mass and can grossly distort expectations when g grows in the tails.

Contributions of this note.

1. We state four baselines—*center-only*, *moment-matched*, *column-tensor*, and *profile–Rosenblatt*—as **approximations to the same measure** on $(z, \mathbf{x})$ (or to a reduced equivalent), with clear bias mechanisms (especially *moment-matched* infinite tails versus compact support along the mean-gradient path).

2. We derive the **tensor-product** reference discretization (*column-tensor*) from Fubini's theorem and record the **exactness class**: tensor products of univariate Gaussian quadrature rules integrate tensor products of polynomials up to the usual univariate degrees—*not* all polynomials of a fixed **total** degree in (z, x_1, x_2); sparse grids (Smolyak) address that different design goal [5].

3. We explain why **Legendre quadrature in z alone** does not solve the problem unless the inner $\mathbb{R}^2$ expectations are treated separately, and why a naive **three-dimensional GL tensor on a box** in (z, x_1, x_2) is **wrong** (Gaussian states are not uniform on a compact box).

4. We present the *profile–Rosenblatt* scheme as **Rosenblatt conditioning** in coordinates aligned with the mean gradient $\mathbf{d}$: outer quadrature for a Gaussian marginal, inner **inverse-CDF** (probability-space GL) for the pill marginal at each outer node.

5. We justify **Brent's method** as inversion of a known monotone CDF F at fixed GL probabilities—a **numerical** step, not part of the physics—and list alternatives (adaptive integration in u, custom quadrature for weight f, Monte Carlo).

6. We frame Chebyshev/rational surrogates by an **operation-count** bottleneck and validate with **interpretable claims** (consistency, structural bias, surrogate fidelity), not timing recipes alone.

Reader's map. Section 2 fixes notation. Section 3 gives a moist (T, q) example. Section 4 answers: *What measure does each quadrature rule encode?* Section 5 defines *center-only*, *moment-matched*, *column-tensor*, and *profile–Rosenblatt*. Sections 6–8 build the pill and quantiles. Sections 9–11 cover surrogates. Section 13 states numerical claims C1–C4. Appendices record stability knobs, optional interpolation, reproducibility, and an example saturation closure.

2 Scope, assumptions, and notation

2.1 Modeling assumptions

We work with a single layer of half-depth $H > 0$ and a bivariate state $\mathbf{x} \in \mathbb{R}^2$. (In moist columns one may take $\mathbf{x} = (T, q)^\top$; the algebra is the same for any two prognostic components coupled through g.) The **layer-mean profile** is linear in z:

$$\mathbb{E}[\mathbf{x} \mid z] = \boldsymbol{\mu} + z\,\mathbf{d}, \qquad z \in [-H/2, H/2], \tag{2}$$

with constant gradient vector $\mathbf{d}$ (in moist coordinates, $\mathbf{d} = (\partial T/\partial z,\, \partial q/\partial z)^\top$). Fluctuations about that mean are **homogeneous Gaussian** turbulence Σ_{turb} independent of z. These assumptions isolate how *vertical structure* of the mean and *nonlinearity* of g interact; they are not a full turbulence or microphysics model.

2.2 Notation

Symbol	Meaning
H	Layer half-depth; $z \in [-H/2, H/2]$
$\boldsymbol{\mu}$	State at layer center ($z = 0$)
$\mathbf{d}$	Gradient of layer-mean profile w.r.t. z (written $\nabla\phi$ in (T, q) formulas below)
Σ_{turb}	SGS covariance (2×2)
N	Quadrature order per axis where a tensor product is used
t_i, w_i	Gauss–Legendre nodes on $[-1, 1]$ and weights
p_i	$(t_i + 1)/2 \in (0, 1)$ (uniform probability nodes for inverse-CDF maps)
M, M^{-1}	Linear map to coordinates along / across $\mathbf{d}$ and its inverse (§6)
Σ_{uv}	$M\,\Sigma_{\text{turb}}\,M^\top$
s_v, $s_{u\|v}$	Marginal and conditional standard deviations in (u, v)
L	Half-width of uniform component along mean path; here $L = H$
η	$s_{u\|v}/L$
F, f, f', f''	CDF/PDF of pill marginal and derivatives (§7)
Discretization names (Section 5):	
center-only	Gaussian SGS at layer center only; no z-integral
moment-matched	Single bivariate Gaussian with $\Sigma_{\text{tot}} = \Sigma_{\text{turb}} + \frac{H^2}{12}\mathbf{d}\mathbf{d}^\top$
column-tensor	Gauss–Legendre in z × mapped GL in $\mathbf{x}$ at each z_k (reference $\mathcal{O}(N^3)$)
profile–Rosenblatt	Rosenblatt factorization along $\mathbf{d}$ with pill inner marginal ($\mathcal{O}(N^2)$)

Path half-width $L = H$. The coordinate u along the mean direction is normalized so that one unit of u corresponds to one unit of physical z along the layer; the uniform factor in the pill has support $[-L/2, L/2] = [-H/2, H/2]$. This is *not* arc length $H\|\mathbf{d}\|$ in state space; Jacobians are absorbed in M.

In Sections 6–12 we write $\nabla\phi$ for $\mathbf{d}$ to match common meteorological notation in (T, q) coordinates.

3 Motivating example: saturation excess in (T, q) space

As a concrete g, consider specific humidity q and temperature T with a prescribed saturation curve $q_{\mathrm{sat}}(T)$ (Appendix C). Define

$$g(\mathbf{x}) = \max\bigl(0,\, q - q_{\mathrm{sat}}(T)\bigr), \qquad \mathbf{x} = (T, q)^{\top}.$$

Then $\overline{G}$ in (1) is a layer-mean positive saturation excess. The nonlinearity is strong (Clausius–Clapeyron-type q_{sat}) and the map is **non-smooth** at the saturation boundary, which stresses quadrature but does not change the measure-theoretic setup.

Gaussian SGS is a **tractable** baseline PDF closure; skewed moisture PDFs and horizontal heterogeneity are outside scope.

The **cost** of a naive tensor discretization with $N_z \sim N$ nodes in z and N^2 nodes in (T, q) per slice is $\mathcal{O}(N^3)$ evaluations of g (and any functions composed with it). The *profile–Rosenblatt* reduction achieves $\mathcal{O}(N^2)$ under the stated linear-mean and homogeneous-covariance assumptions.

4 Gaussian quadrature and the measures it encodes

4.1 Univariate Gaussian quadrature: what "optimal" means

Gaussian quadrature approximates

$$\int_a^b w(t)\, h(t)\, dt \approx \sum_{i=1}^{N} W_i\, h(t_i)$$

by choosing nodes $\{t_i\}$ and weights $\{W_i\}$ so that the rule is **exact** whenever h is a polynomial of degree $\leq 2N - 1$, with respect to the **positive weight** w on (a, b) [2, 3]. Different choices of (a, b, w) yield different classical rules. There is no single rule that is "optimal for all integrals"; mismatches between w and the actual density induce systematic node placement error.

4.2 Gauss–Legendre = Gaussian quadrature for uniform weight on a bounded interval

On $[-1, 1]$ with $w(t) = 1$, the orthogonal polynomials are Legendre polynomials; the resulting nodes $\{t_i\}$ and weights $\{w_i\}$ satisfy

$$\int_{-1}^{1} p(t)\, dt = \sum_{i=1}^{N} w_i\, p(t_i) \tag{3}$$

for all polynomials p of degree $\leq 2N - 1$. Any affine map from a finite interval $[a, b]$ to $[-1, 1]$ pushes Lebesgue measure forward to a scaled uniform measure; Gauss–Legendre on $[-1, 1]$ is therefore the natural N-point rule for **uniform** averaging over a compact domain when the integrand is sufficiently smooth in the mapped variable.

Outer integral in z. The layer average uses $\frac{1}{H}\int_{-H/2}^{H/2} F(z)\,dz$ with flat measure. The map $z = (H/2)\,t$, $t \in [-1,1]$, gives

$$\frac{1}{H}\int_{-H/2}^{H/2} F(z)\,dz = \frac{1}{2}\int_{-1}^{1} \tilde{F}(t)\,dt, \qquad \tilde{F}(t) = F\big((H/2)t\big).$$

Gauss–Legendre in t is the standard high-order choice when $\tilde{F}$ is smooth. If g induces kinks in $z \mapsto \mathbb{E}[g(\mathbf{X}(z))]$, lower-order or adaptive z-rules may be preferable—a modeling regularity question separate from the inner Gaussian structure.

4.3 Gauss–Hermite = Gaussian quadrature for Gaussian weight on the line

Physicists' Gauss–Hermite nodes $\{x_i\}$ and weights $\{W_i\}$ integrate

$$\int_{-\infty}^{\infty} e^{-x^2}\,\tilde{h}(x)\,dx \approx \sum_i W_i\,\tilde{h}(x_i)$$

exactly for polynomials $\tilde{h}$ of degree $\leq 2N-1$. With $z = \sqrt{2}\,x$,

$$\int_{-\infty}^{\infty} e^{-x^2}\,\tilde{h}(x)\,dx = \int_{-\infty}^{\infty} \phi(z)\,h(z)\,dz$$

for $h(z) = \tilde{h}(z/\sqrt{2})$ and ϕ the standard normal PDF. Thus GH implements Gaussian quadrature for the **normal weight** $\phi(z)\,dz$ on $\mathbb{R}$ [3]. Tensor products of GH rules in two coordinates integrate tensor polynomials against **bivariate standard normal** weight.

4.4 Mismatched rules: why Hermite nodes are wrong for the pill marginal

Along the inner axis of the *profile–Rosenblatt* scheme (at fixed outer v), the relevant 1D law is the **pill**: convolution of a Gaussian with a uniform on $[-L/2, L/2]$. Its density f_{pill} is flat in the center with Gaussian tails—not proportional to ϕ. Gauss–Hermite quadrature is constructed to be exact for polynomials multiplied by ϕ; using GH as if the marginal were standard normal allocates nodes as if mass were Gaussian throughout, missing the uniform core. The correct repair is either (i) quadrature with respect to the true weight f_{pill} (non-classical: requires orthogonal polynomials for f_{pill} or adaptive methods), or (ii) the **inverse-CDF** construction of §4.5, which applies GL on $[0,1]$ to $\int_0^1 h(F^{-1}(p))\,dp$.

4.5 From uniform probability to expectations: the inverse-CDF identity

Let X be a scalar with CDF F that is strictly increasing on the support where F^{-1} exists. If $U \sim \text{Unif}(0,1)$, then $F^{-1}(U)$ has law F. Hence for any integrable h,

$$\mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x)\,dF(x) = \int_0^1 h(F^{-1}(p))\,dp. \tag{4}$$

The second integral is an expectation under **uniform** p. Gauss–Legendre nodes $t_i \in (-1,1)$ with weights w_i integrate polynomials on $[-1,1]$ exactly; the affine map $p_i = (t_i+1)/2$, weights $w_i/2$, integrates polynomials in p on $[0,1]$. Therefore

$$\mathbb{E}[h(X)] \approx \sum_{i=1}^{N} \frac{w_i}{2}\,h\big(F^{-1}(p_i)\big)$$

is **Gaussian quadrature for uniform measure on** $[0,1]$ **applied to the composite function** $p \mapsto h(F^{-1}(p))$. This is the source of all "$p_i = (t_i+1)/2$" factors in the pill inner rule. It is *not* Gauss–Hermite on the x-axis; the Hermite construction would target $\int h(x)\phi(x)\,dx$ instead of $\int h(x) f_{\text{pill}}(x)\,dx$.

4.6 Probability-transform (mapped) Gauss–Legendre for $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ in $\mathbb{R}^2$

For $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ with $\Sigma = LL^\top$, let $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_2)$. Then $\mathbf{X} = \boldsymbol{\mu} + L\mathbf{Z}$. For each component, $\mathbb{E}[h_j(Z_j)] = \int_0^1 h_j(\Phi^{-1}(p))\, dp$ as in (4). A **tensor product** in two independent standard normals yields nodes $\mathbf{z}_{ij} = (\Phi^{-1}(p_i), \Phi^{-1}(p_j))$ with weights $(w_i w_j)/4$, then $\mathbf{x}_{ij} = \boldsymbol{\mu} + L\mathbf{z}_{ij}$. This is the mapped GL construction used inside *center-only*, *moment-matched*, and *column-tensor* at fixed z.

Optional tail stretching ($\alpha \geq 1$). For **diagnostics** only—comparing a closure that places mass in non-physical Gaussian tails (e.g. *moment-matched*) to a more faithful scheme—one may evaluate g at $\boldsymbol{\mu} + L(\alpha\mathbf{z})$ while retaining quadrature weights designed for $\alpha = 1$. The Jacobian and likelihood ratio produce the correction (stated here without re-deriving):

$$w_{ij}^{(\mathrm{GL})} = \frac{w_i w_j}{4}\, \alpha^2 \exp\Big(-\tfrac{1}{2}\big(1 - \alpha^{-2}\big)\, d\Big), \qquad d = \|\mathbf{z}_{ij}\|^2. \tag{5}$$

$\alpha = 1$ is the unbiased probability-transform rule; $\alpha > 1$ is **intentionally biased** sampling for experiments.

```
t, w = leggauss(N)
z1, z2 = meshgrid(t, t); w1, w2 = meshgrid(w, w)
Z_std = alpha * sqrt(2) * erfinv(vstack([z1.ravel(), z2.ravel()]))
phys = mu + Lchol @ Z_std
dist_sq = sum(Z_std**2, axis=0)
weights = (w1*w2).ravel()/4 * alpha**2 * exp(-0.5*(1 - 1/alpha**2) *
    dist_sq)
```

4.7 Tensor products, Fubini, and the column-tensor discretization

The target (1) is an iterated integral:

$$\overline{G} = \int_{-H/2}^{H/2} \left(\int_{\mathbb{R}^2} g(\mathbf{x})\, \phi_z(\mathbf{x})\, d\mathbf{x} \right) \frac{dz}{H}.$$

Fubini's theorem applies under standard integrability conditions on g and Gaussian tails. The outer measure is **uniform** on $[-H/2, H/2]$; the inner measure at each z is **Gaussian** on $\mathbb{R}^2$.

Tensor-product quadrature. If the outer z-integral is approximated by a N_z-point GL rule after $z = (H/2)t$, and each inner integral by an N^2 mapped GL tensor in $\mathbf{x}$ as above, the combined rule is a **tensor product** of univariate Gaussian quadrature rules on the **product measure** (uniform $\times$ bivariate normal). Its standard exactness statement is: the combined rule integrates any function that is a **finite sum of products** of polynomials of degrees (d_t, d_1, d_2) in (t, x_1, x_2) up to the univariate limits $d_t \leq 2N_z - 1$, $d_1, d_2 \leq 2N - 1$ *in the Gaussian-transformed coordinates*. It does **not** integrate all polynomials of total degree $\leq D$ in (t, x_1, x_2) for modest D without increasing N sharply; sparse Smolyak constructions reduce cost for smooth isotropic integrands on $\mathbb{R}^d$ [5] but are not implemented here.

4.8 Why Legendre quadrature on the layer alone is insufficient

Two common confusions are worth separating.

(A) Only discretizing z. Replacing $\frac{1}{H}\int F(z)\, dz$ by a GL sum requires values $F(z_k) \approx \mathbb{E}[g(\mathbf{X}(z_k))]$. If the inner expectations are *not* computed—if one plugs a single representative $\mathbf{x}$ per cell—the result is not (1). Quadrature in z addresses only the **outer** average once $F(z)$ is available.

(B) A three-dimensional tensor Gauss–Legendre box in (z, x_1, x_2). If one mapped (z, x_1, x_2) to $[-1, 1]^3$ and applied GL in each coordinate, one would implicitly integrate against a **uniform** density on a **compact parallelepiped**. But $\mathbf{X}(z)$ is **unbounded** with Gaussian tails at every z; the correct inner measure is not uniform on any box. The valid 3D structure is **mixed**: compact uniform z (after normalization) $\times$ non-compact Gaussian in $\mathbf{x}$ at each z (*column-tensor*), or, after *profile–Rosenblatt*, uniform-in-z folded into a pill times a Gaussian in v.

4.9 Rosenblatt conditioning

For a bivariate density f_{UV}, always $f_{UV}(u, v) = f_V(v)\, f_{U|V}(u \mid v)$. A consistent product quadrature uses a rule for f_V in v and, **for each** v_j, a rule for $f_{U|V}(\cdot \mid v_j)$ in u. Rotating (x_1, x_2) to decorrelate without conditioning and then tensoring **marginal** 1D rules for u and v is generally **incorrect**: the joint iso-density curves are not aligned with the coordinate grid implied by independent marginals. Rosenblatt's transform [4] systematizes monotone triangular maps that factor a multivariate law into univariate conditionals; the *profile–Rosenblatt* construction is the instance aligned with the mean gradient $\mathbf{d}$ in which the inner marginal becomes the pill.

5 Layer-mean objective and baseline approximations

5.1 Integral statement (moist specialization)

For the motivating g,

$$\overline{q_c} = \frac{1}{H} \int_{-H/2}^{H/2} \iint_{\mathbb{R}^2} \max\bigl(0,\, q - q_{\mathrm{sat}}(T)\bigr)\, \mathcal{N}\bigl(\mathbf{x} \mid \boldsymbol{\mu} + z\mathbf{d},\, \Sigma_{\mathrm{turb}}\bigr)\, dT\, dq\, dz. \tag{6}$$

This is (1) with $\mathbf{d} = \nabla\phi$ in (T, q) coordinates. Any consistent scheme must respect both the z-dependence of the Gaussian mean and Σ_{turb}.

5.2 Center-only: collapse the layer to the center

Approximate measure. Replace the z-integral by sampling only $z = 0$: use $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma_{\mathrm{turb}})$ and $\overline{q_c} \approx \mathbb{E}[g(\mathbf{X})]$ with the same N^2 mapped GL tensor as in *moment-matched/column-tensor* inner rules.

Cost. $\mathcal{O}(N^2)$ evaluations of g per layer.

Bias. All vertical variation of the layer-mean profile is discarded. If the segment $\{\boldsymbol{\mu} + z\mathbf{d}\}$ crosses a manifold where g activates (e.g. saturation), mass that should arise at intermediate z is missing; $\overline{q_c}$ is typically *biased low* relative to (6).

5.3 Moment-matched: single Gaussian with total variance

Approximate measure. Use $\mathcal{N}(\boldsymbol{\mu}, \Sigma_{\mathrm{tot}})$ with

$$\Sigma_{\mathrm{tot}} = \Sigma_{\mathrm{turb}} + \frac{H^2}{12}\, \mathbf{d}\, \mathbf{d}^\top, \tag{7}$$

matching the second moments of $\mathbf{x}$ induced by uniform z and independent Gaussian SGS *if* one ignored the compact-support structure along the mean direction.

Cost. $\mathcal{O}(N^2)$.

Bias / diagnostic value. Averaging z over a finite interval induces, in coordinates aligned with $\mathbf{d}$, a **uniform** component along the mean-gradient direction convolved with Gaussian SGS—compact support in that direction at fixed transverse v. *moment-matched* replaces that by an **infinite-tailed** Gaussian with matched low-order moments. Tail quadrature nodes then

probe (T, q) regions where the true averaged measure has negligible mass; if g grows rapidly there (cold tails of q_{sat}), *moment-matched* can **overestimate** (6). It remains a useful diagnostic of moment-matching closures, not a faithful substitute for *column-tensor* or *profile–Rosenblatt*.

5.4 Column-tensor: tensor product in z and (T, q)

Approximate measure. Discretize the outer z-average with GL as in §4.7: nodes $t_k^{(z)}$, $z_k = (H/2)t_k^{(z)}$, weights $w_k^{(z)}/2$. At each k, approximate $\mathbb{E}[g(\mathbf{X}_k)]$ with $\mathbf{X}_k \sim \mathcal{N}(\boldsymbol{\mu} + z_k\mathbf{d}, \Sigma_{\text{turb}})$ by an N^2 mapped GL rule. Accumulate $\sum_k (w_k^{(z)}/2)\, I_k$.

Cost. $\mathcal{O}(N_z N^2)$; with $N_z \sim N$, cost $\mathcal{O}(N^3)$.

Role. *column-tensor* is the direct numerical realization of Fubini + tensor quadrature before analytic reduction; it serves as a reference when N_z is large.

Algorithm 1 Column-tensor (tensor-z) estimate of $\overline{q_c}$

```
Input: H, μ, d, Σ_turb, orders N_z, N
(t_k^(z), w_k^(z)) ← Legendre(N_z) on [−1, 1]
q_sum ← 0
for k = 1 to N_z do
    z_k ← (H/2) t_k^(z),   W_k ← w_k^(z)/2
    μ_k ← μ + z_k d
    {(x_ℓ^(k), W_ℓ^(2))}_{ℓ=1}^{N^2} ← MappedGL2D(μ_k, Σ_turb, N, α)
    I_k ← Σ_ℓ W_ℓ^(2) max(0, x_ℓ^(q) − q_sat(x_ℓ^(T)))
    q_sum ← q_sum + W_k I_k
end for

return q_sum
```

5.5 Profile–Rosenblatt: quadrature along the mean gradient

Reduced representation. Under linear mean in z and homogeneous Σ_{turb}, the z-average and bivariate Gaussian can be combined so that $\overline{q_c}$ becomes a double sum: an outer quadrature in a Gaussian coordinate v transverse to $\mathbf{d}$ in state space, and an inner quadrature in u along $\mathbf{d}$ against the **pill** marginal at each v (Sections 6–12).

Cost. $\mathcal{O}(N^2)$ evaluations of g for equal orders N.

Role. *profile–Rosenblatt* is the accurate $\mathcal{O}(N^2)$ target under the stated assumptions.

6 Gradient-aligned coordinates: the maps M, M^{-1}, and path length

6.1 Construction of M

Let $\nabla\phi \equiv \mathbf{d}$ be the gradient of the mean profile. Row vectors

$$\mathbf{u}_{\text{row}} = \frac{\nabla\phi}{\|\nabla\phi\|^2}, \qquad \mathbf{v}_{\text{row}} = \big(-(\nabla\phi)_q,\, (\nabla\phi)_T\big)^\top \tag{8}$$

define coordinates $u = \mathbf{u}_{\text{row}}\mathbf{x}$ and $v = \mathbf{v}_{\text{row}}\mathbf{x}$ for departures from $\boldsymbol{\mu}$. Stacking rows,

$$M = \begin{bmatrix} \mathbf{u}_{\text{row}}^\top \\ \mathbf{v}_{\text{row}}^\top \end{bmatrix}, \qquad M^{-1} = \begin{bmatrix} \nabla\phi & \mathbf{v}_{\text{row}}/\|\nabla\phi\|^2 \end{bmatrix}. \tag{9}$$

The first row is scaled so that a unit change in u matches one unit of physical z along the layer when the profile is linear.

6.2 Conditional Gaussian parameters

Let $\Sigma_{uv} = M\,\Sigma_{\text{turb}}\,M^\top = \begin{pmatrix} s_1^2 & s_{12} \\ s_{12} & s_2^2 \end{pmatrix}$. Then $s_v = \sqrt{s_2^2}$ and

$$s_{u|v} = \sqrt{\max\Big(s_1^2 - \frac{s_{12}^2}{\max(s_2^2, \varepsilon)},\, 0\Big)}, \qquad \mu_{u|v}(v) = \frac{s_{12}}{\max(s_2^2, \varepsilon)}\, v. \tag{10}$$

Map back:

$$\mathbf{x} = \boldsymbol{\mu} + M^{-1} \begin{pmatrix} u + \mu_{u|v}(v) \\ v \end{pmatrix}. \tag{11}$$

6.3 Uniform path half-width $L = H$ and degenerate gradients

$L = H$ identifies the uniform factor with $z \in [-H/2, H/2]$. If $\|\nabla\phi\| < \varepsilon$, use the *center-only* scheme at $\boldsymbol{\mu}$ instead of forcing *profile–Rosenblatt*.

7 The pill marginal along u at fixed v

7.1 Where the uniform factor comes from

At fixed v, SGS contributes Gaussian uncertainty along u with variance $s_{u|v}^2$. The layer average over uniform z adds an independent uniform displacement of width H along the same physical direction (after the normalization of M). Their sum is the **pill**; its density is trapezoidal in the center with Gaussian tails.

7.2 CDF, PDF, and derivatives

Let $s \equiv s_{u|v}$, $Y \sim \mathcal{N}(0, s^2)$, $U_{\text{un}} \sim \text{Unif}([-L/2, L/2])$ independent, and $X = Y + U_{\text{un}}$ (centering via $\mu_{u|v}(v)$ handled in Algorithm 5). Then

$$F(x) = \frac{s}{L}\left[\mathcal{I}\Big(\frac{x + L/2}{s}\Big) - \mathcal{I}\Big(\frac{x - L/2}{s}\Big)\right], \tag{12}$$

with $\mathcal{I}(y) = y\Phi(y) + \phi(y)$. With $u_1 = (x + L/2)/s$, $u_2 = (x - L/2)/s$,

$$f(x) = F'(x) = \frac{1}{2L}\left[\text{erf}\Big(\frac{u_1}{\sqrt{2}}\Big) - \text{erf}\Big(\frac{u_2}{\sqrt{2}}\Big)\right], \tag{13}$$

$$f'(x) = \frac{1}{Ls\sqrt{2\pi}}\left[e^{-u_1^2/2} - e^{-u_2^2/2}\right], \tag{14}$$

$$f''(x) = \frac{1}{Ls^2\sqrt{2\pi}}\left[-u_1 e^{-u_1^2/2} + u_2 e^{-u_2^2/2}\right]. \tag{15}$$

7.3 Inner quadrature: Gauss–Legendre in p (probability space)

At fixed v, the inner expectation is $\int_{\mathbb{R}} g_{\text{cond}}(u)\, f(u)\, du$ for some conditional integrand g_{cond} composed with the map from (u, v) to physical $\mathbf{x}$. By (4),

$$\int_{\mathbb{R}} g_{\text{cond}}(u)\, f(u)\, du = \int_0^1 g_{\text{cond}}(F^{-1}(p))\, dp.$$

The integrand on $[0, 1]$ is a **composition** of $g_{\text{cond}} \circ F^{-1}$. Gauss–Legendre on $[-1, 1]$ mapped to $[0, 1]$ ($p_i = (t_i + 1)/2$, weights $w_i/2$) is Gaussian quadrature for **uniform** p. Nodes $u_i = F^{-1}(p_i)$ are **quantiles** of the pill; this is standard *inverse-CDF* or *quantile* quadrature [3]. **Contrast:** Gauss–Hermite quadrature in u targets $\int \cdot\, \phi(u)\, du$, not $\int \cdot\, f_{\text{pill}}(u)\, du$, and is the wrong tool for this marginal.

8 Why root-finding appears, and Brent's method

8.1 Root-finding is not "physics"

The identity (4) requires $u_i = F^{-1}(p_i)$. For the pill, F is explicit (12) but F^{-1} is not elementary. **Alternatives** to bracketing root-finding include: (i) adaptive quadrature of $\int g_{\text{cond}}(u) f(u)\, du$ without fixed quantile nodes; (ii) constructing Gaussian quadrature abscissas for the non-classical weight f (solve the orthogonal-polynomial / moment problem—substantial preprocessing); (iii) Monte Carlo or quasi-Monte Carlo in u; (iv) Newton/Halley iterations on $F(u) - p_i$ given cheap f, f' (Section 9). The **design choice** here is fixed GL probabilities $\{p_i\}$ and quantile nodes; then u_i are defined by $F(u_i) = p_i$.

8.2 Brent's method

Brent's method [1] robustly solves $F(x) - p_i = 0$ for monotone F. **Bracket:** $[-(L/2+6s),\ L/2+6s]$ for interior p_i when $s > 0$. **Degenerate** $s \to 0$**:** $F^{-1}(p_i) = t_i L/2$ with GL weights $w_i/2$ on the uniform limit.

Algorithm 2 Brent quantiles for pill marginal

```
Input: L > 0, s ≥ 0, Legendre {t_i, w_i}_{i=1}^N
if s < ε then
    u_i ← t_i L/2, W_i ← w_i/2

    return {u_i}, {W_i}
end if
for i = 1 to N do
    p_i ← (t_i + 1)/2
    u_i ← Brent(F(·) − p_i, −(L/2 + 6s), L/2 + 6s)
    W_i ← w_i/2
end for
return {u_i}, {W_i}
```

8.3 Complexity and the surrogate motivation

For each outer node v_j, *profile–Rosenblatt* needs $\{u_i\}$ at $s = s_{u|v}(v_j)$: $\mathcal{O}(N)$ Brent solves per slice, $\mathcal{O}(N^2)$ per layer if done naively. Each Brent iteration evaluates F (and possibly f, f' for Halley). Surrogates replace most Brent work by evaluating a polynomial or rational in $\eta = s/L$ (Section 9).

9 Surrogate pill inversions

The *profile–Rosenblatt* bottleneck is repeated inversion of the **same functional form** F at many $(L, s, \{p_i\})$ with $L = H$ fixed per layer but $s = s_{u|v}(v_j)$ varying across outer nodes. Empirically, u_i varies smoothly in $\log_{10} \eta$ on fitted ranges; this **heuristic smoothness** motivates fitting $u_i(\eta)$ offline with Brent as reference, then evaluating fits at runtime [1].

Surrogates must be validated against Brent (claim C3, §13).

9.1 Hybrid uniform–Gaussian blending (experimental)

For $\eta = s/L \ll 1$, quantiles are near uniform; for $\eta \gg 1$, near Gaussian. Sigmoid blends can seed iterations but lack error guarantees unless paired with residual checks.

9.2 Halley iteration on $F(x) = p_i$

$$x \leftarrow x_0 - \frac{2f(x_0)\,(F(x_0) - p_i)}{2f(x_0)^2 - (F(x_0) - p_i)\,f'(x_0)}. \tag{16}$$

9.3 Logistic seed + Householder step

$$\mathrm{Var}_{\mathrm{tot}} = s^2 + \frac{L^2}{12}, \quad S_{\log} = \frac{\sqrt{3\,\mathrm{Var}_{\mathrm{tot}}}}{\pi}, \quad x_0 = S_{\log} \ln\left(\frac{p_i}{1 - p_i}\right). \tag{17}$$

With $h = (p_i - F(x_0))/f(x_0)$,

$$x_{\mathrm{final}} = x_0 + h\left[1 + \frac{h}{2}\frac{f'(x_0)}{f(x_0)} + \frac{h^2}{6}\left(\left(\frac{f'(x_0)}{f(x_0)}\right)^2 + \frac{f''(x_0)}{f(x_0)}\right)\right]. \tag{18}$$

10 Chebyshev approximation in $\log_{10}\eta$

10.1 Rationale and operation count

Offline: for each η_k on a log grid, run Algorithm 2 once ($\mathcal{O}(N)$ Brent solves $\times\, n_\eta$ points). Online: evaluate degree-D Chebyshev sums—$\mathcal{O}(ND)$ flops per layer slice versus $\mathcal{O}(N)$ iterative solves $\times$ cost per iteration for Brent.

10.2 Training grid and Brent truth

$$\eta_k = 10^{a + (b-a)\frac{k-1}{n_\eta - 1}}, \quad a = \log_{10}\eta_{\min},\ b = \log_{10}\eta_{\max}. \tag{19}$$

Set $\sigma_k = \eta_k L_{\mathrm{train}}$; store $u_{k,i}^{\mathrm{true}}$.

10.3 Affine map and symmetry

Let $\ell_k = \log_{10}\eta_k$, $\ell_{\min} = \min_k \ell_k$, $\ell_{\max} = \max_k \ell_k$, and $\tau_k = 2(\ell_k - \ell_{\min})/(\ell_{\max} - \ell_{\min}) - 1$. The pill quantiles are odd-symmetric about the mean for symmetric p_i; fit coefficients only for indices $i \geq i_{\mathrm{start}} = \lfloor (N+1)/2 \rfloor$ (1-based) and recover the other half by $u_{N+1-i} = -u_i$.

10.4 Runtime evaluation and offline validation

At runtime: form $\eta = s/L_{\mathrm{path}}$, map $\ell = \log_{10}\eta$ to τ, evaluate Chebyshev sums for u_i^+, mirror to u_i^-, keep weights $w_i/2$. On a denser $\{\tilde{\eta}_m\}$ than the training grid, compare Brent and Chebyshev u_i and report $\max_{m,i}$ absolute error.

Algorithm 3 Offline Chebyshev build for pill quantiles

```
Choose N, L, η_min, η_max, n_η, degree D
Build {η_k} log-uniform; compute truth rows u_k^true
Form τ_k from log10 η_k
for i = i_start to N do
    y_k ← u_{k,i}^true / L_train
    c^(i) ← ChebyshevLS({τ_k}, {y_k}, D)
end for

return {c^(i)}, meta
```

11 Rational surrogate in η

Rational forms can approximate $u_i(\eta)$ with fewer parameters than high-degree polynomials when u_i is gently curved in η on the training range; spurious poles where $Q(\eta) \approx 0$ require fallbacks to Brent or another surrogate.

11.1 Template and training

For each fitted node index,

$$r(\eta) = \frac{a_0 + a_1\eta + a_2\eta^2 + a_3\eta^3}{1 + b_1\eta + b_2\eta^2}. \tag{20}$$

Train on logarithmically spaced $\{\eta_k\}$ with Brent truth from Algorithm 2; fit $(a_0, \ldots, b_2)$ by nonlinear least squares (e.g. Levenberg–Marquardt). Positivity constraints on coefficients are an engineering stabilizer, not a physical statement.

11.2 Failure modes

Non-convergence of the fit, or $|1+b_1\eta+b_2\eta^2|$ small at runtime: fall back to Brent or Householder.

Algorithm 4 Rational pill quantiles (cached)

```
Input: N, L_path, s, cache C_N
η ← s / max(L_path, ε)
if η < ε then
    return uniform quantiles on [−L_path/2, L_path/2]
end if
Evaluate r_j(η) for each stored coefficient vector
Assemble symmetric node vector; weights w_i/2
return nodes, weights
```

12 Profile–Rosenblatt assembly for $\overline{q_c}$

Algorithm 5 Profile–Rosenblatt quadrature for $\overline{q_c}$ (schematic)

```
if ‖∇φ‖ < ε then
    return center-only estimate (2D rule at μ)
end if
Build M, M^{-1}, Σ_uv; extract s_v, s_12, s_2^2, s_1^2
Form 1D Gaussian rule {v_j, W_{v,j}} on N(0, s_v^2)
for j = 1 to N do
    s ← s_{u|v} from (s_1^2, s_2^2, s_12, v_j)
    {u_i, W_{u,i}} ← PillQuantiles(N, L, s) {Brent or surrogate}
    μ_u ← (s_12 / max(s_2^2, ε)) v_j
    for i = 1 to N do
        x_ij ← μ + M^{-1}(u_i + μ_u, v_j)^⊤
        f_ij ← max(0, x_ij^(q) − q_sat(x_ij^(T)))
    end for
end for

return Σ_ij f_ij W_{u,i} W_{v,j}
```

13 Numerical study: claims, metrics, interpretation

This section states **what** each numerical check tests and **how to read** failure. Procedural detail for reproducing timings belongs in Appendix A.

13.1 C1: Consistency of discretizations

Claim. With sufficiently large orders, *column-tensor* (large N_z) and *profile–Rosenblatt* should agree within tolerances set by residual quadrature error if both are implemented correctly. For each method, increasing N (and N_z for *column-tensor*) should yield a plateau; comparing methods before this plateau can confuse quadrature error with structural bias.

Failure modes. Systematic disagreement suggests a bug, mismatched α tail stretching, or insufficient N_z / N.

13.2 C2: Structural bias of the moment-matched Gaussian

Claim. At large N, *moment-matched* remains biased relative to *column-tensor*/*profile–Rosenblatt* when g amplifies tail regions where *moment-matched* places mass but the true averaged measure does not (§5).

Interpretation. This is not "quadrature error" in the usual sense; changing N cannot remove the wrong measure.

13.3 C3: Surrogate fidelity and total error against a reference

Let ε_{abs} be a tiny positive floor and $\overline{q_c}^{\text{ref}}$ a high-resolution reference (§13.5). Define (subscript PR denotes *profile–Rosenblatt*; B / R denote Brent vs. rational pill inversion)

$$e_{\text{PR,B}}(N) = \frac{\left|\overline{q_c}^{\text{PR,Brent}}(N) - \overline{q_c}^{\text{ref}}\right|}{\max(|\overline{q_c}^{\text{ref}}|, \varepsilon_{\text{abs}})}, \tag{21}$$

$$e_{\text{PR,R}}(N) = \frac{\left|\overline{q_c}^{\text{PR,rational}}(N) - \overline{q_c}^{\text{ref}}\right|}{\max(|\overline{q_c}^{\text{ref}}|, \varepsilon_{\text{abs}})}, \tag{22}$$

$$e_{\text{pill}}(N) = \frac{\left|\overline{q_c}^{\text{PR,rational}}(N) - \overline{q_c}^{\text{PR,Brent}}(N)\right|}{\max(|\overline{q_c}^{\text{PR,Brent}}(N)|, \varepsilon_{\text{abs}})}. \tag{23}$$

Interpretation. $e_{\text{PR,B}}$ is total quadrature error of Brent-based *profile–Rosenblatt* against the reference (includes reference bias). $e_{\text{PR,R}}$ adds surrogate error in the same norm. e_{pill} **isolates** inverse-CDF surrogate error with all other *profile–Rosenblatt* choices identical. Small pointwise errors in u_i can still move $\overline{q_c}$ substantially if g is sharply varying (e.g. saturation boundary); report $\max_{k,i} |u_i^{\text{Brent}} - u_i^{\text{approx}}|$ on an η grid to catch unstable fits.

13.4 C4: Complexity reference values versus surrogates

Claim. Brent-based *profile–Rosenblatt* performs $\mathcal{O}(N)$ root-finds per outer node, each requiring several evaluations of F; surrogate *profile–Rosenblatt* replaces those by $\mathcal{O}(ND)$ or $\mathcal{O}(N)$ rational evaluations with fixed coefficient tables.

Interpretation. Report operation counts or mean timings *as evidence for this scaling*, not as an end in itself; see Appendix A for a reproducibility template.

13.5 High-resolution reference $\overline{q_c}^{\text{ref}}$

No closed form is available for realistic q_{sat}. A dense uniform grid in z (e.g. $n_z = 250$) with high-order inner mapped GL in (T, q) at each z (e.g. $N_{\text{2d}} = 50$) provides a numerical reference; residual bias from finite n_z and N_{2d} should be driven below the method differences among the four schemes and surrogate tests.

13.6 Optional graphics

Plots of nodes in (T, q) with weights illustrate measure mismatch for *moment-matched* versus *profile–Rosenblatt*; they support intuition but are not normative.

14 Worked micro-example ($N = 2$ pill quantiles)

For $N = 2$, $t_i = \pm 1/\sqrt{3}$, $p_i = (1 \pm 1/\sqrt{3})/2$, weights $w_i/2 = 1/2$. Brent solves $F(u_i) = p_i$; this is the optimal GL pair on $[0, 1]$ composed with F^{-1}.

A Reproducibility protocol for timing (supports C4)

Warm up compiled code once; repeat many calls; discard outliers; report mean and standard deviation of wall time per call for Brent-based versus surrogate quantile construction at fixed (L, s, N). Relate observed ratios to the expected dominance of iterative F evaluations versus polynomial evaluation.

B Optional implementation crosswalk

One atmosphere-model codebase exposes helpers such as `gauss_hermite`, `gauss_legendre_01`, and gradient-aware SGS drivers; the full *profile–Rosenblatt* pipeline (gradient-aligned map M, pill quantiles, Brent reference, cached surrogates) may not be complete in any given repository version. Porting order: implement Algorithm 2; validate; serialize surrogate tables; implement Algorithm 5.

C Example saturation closure for reproducibility

The following Tetens-type closure is **one** choice for cross-model comparisons; operational moisture tables may differ.

```
T_celsius = T_kelvin - 273.15
e_s = 6.1094 * exp((17.625 * T_celsius) / (T_celsius + 243.04))
q_sat = 0.622 * e_s / (P_hPa - 0.378 * e_s)
```

Use the same closure when comparing to published numbers, or document shifts from your production saturation formulation.

D Numerical stability knobs

- Cholesky jitter: $\Sigma + \varepsilon I$ before factorization.
- Denominators: $\max(s_2^2, \varepsilon)$, $\max(L, \varepsilon)$, $\max(f(x), \varepsilon)$.
- Brent brackets scaled with s.

E Barycentric Floater–Hormann interpolation (optional)

$$r(\xi) = \frac{\sum_{k=0}^{M} \frac{w_k}{\xi - \xi_k} y_k}{\sum_{k=0}^{M} \frac{w_k}{\xi - \xi_k}}. \tag{24}$$

See [6].

References

[1] R. P. Brent, *Algorithms for Minimization without Derivatives*, Prentice-Hall, 1973.

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[6] M. S. Floater and K. Hormann, Barycentric rational interpolation with no poles and high rates of approximation, *Numer. Math.* **114** (2009), 1–25.